

$\frac{1}{2}U(X) + \frac{1}{2}U(Y)$ lies on the diagonal line connecting $U(X)$ and $U(Y)$ and is shown by the triangle. Utility is higher for the certain outcome than for the random amount.⁶

$$U\left(\frac{1}{2}X + \frac{1}{2}Y\right) \geq \frac{1}{2}U(X) + \frac{1}{2}U(Y).$$

The greater the difference between the star and the circle in Figure 2.4, the more risk averse the investor. Put another way, the more risk averse the asset owner becomes, the more she wants the sure thing. The more concave the utility function, the more risk averse the investor.

In the special case that the investor is *risk seeking*, the utility function becomes convex rather than concave. If an investor is *risk neutral* the utility function is linear. These cases are rare; most investors are risk averse.

2.3. HOW RISK AVERSE ARE YOU?

Risk aversion can change as wealth changes, and for most individuals risk aversion decreases as wealth increases—individuals generally take on more risk as they become more financially secure. For this reason, I will talk about (relative) risk aversion, where risk aversion is measured relative to wealth, and for short hand I will drop the word “relative” from now on.⁷

CRRA utility takes the following form:

$$U(W) = \frac{W^{1-\gamma}}{1-\gamma}, \quad (2.1)$$

where γ is the asset owner’s risk aversion coefficient. CRRA utility is widely used in portfolio choice theory. A very attractive property of CRRA utility is that it leads to portfolio weights that do not depend on wealth: it doesn’t matter whether one is managing \$10 million or \$100 million—the technology of asset management is exactly the same—and this is the beauty of the scalability of the wealth management industry, at least for the fund manager. (This property is called *wealth homogeneity*). Under CRRA utility, risk aversion is the same for all levels of wealth—so the advantage for this utility function is also an inherent disadvantage, because investors are generally more risk averse over losses than over gains (see section 4). But it is convenient that we can summarize all behavior with just one risk aversion parameter. With this caveat in mind, what risk aversion levels do investors have?

⁶ Or $U(E[X]) \geq E[U(X)]$, which is a consequence of Jensen’s inequality for the concave function.

⁷ Formally, there is the absolute risk aversion coefficient introduced by Arrow (1971) and Pratt (1964) defined as $-U''(W)/U'(W)$ and the relative risk aversion coefficient defined as $-WU''(W)/U'(W)$. Note these definitions of risk aversions are invariant to affine transformations just as utility ordering is itself invariant to affine transformations. Levy’s (1994) subjects exhibited decreasing absolute risk aversion as they become wealthier and approximately CRRA.